General Biology 1

First Semester

Supplementary Materials

Module 1

The Cell Theory

Raymart F. Redita
Author

What I Know

Directions: Read each situation carefully and choose the best answer.

1. A scientist examined a thin slice of cork using a microscope and observed small box-like structures which he called “cells.” Who made this observation?
   a. Antonie van Leeuwenhoek
   b. Rudolf Virchow
   c. Robert Hooke
   d. Matthias Schleiden

2. A student observed bacteria moving under a microscope. The teacher explained that the first scientist who observed living microorganisms used a powerful single-lens microscope. Who was this scientist?
   a. Robert Hooke
   b. Matthias Schleiden
   c. Antonie van Leeuwenhoek
   d. Rudolf Virchow

3. During a biology discussion, a teacher said that all plants are composed of cells. Which scientist first proposed this idea?
   a. Antonie van Leeuwenhoek
   b. Matthias Schleiden
   c. Rudolf Virchow
   d. Robert Hooke

4. A researcher concluded that new cells are formed from existing cells during cell division. Which scientist proposed this principle?
   a. Rudolf Virchow
   b. Robert Hooke
   c. Matthias Schleiden
   d. Antonie van Leeuwenhoek

5. A class is creating a timeline of discoveries that led to the development of cell theory. Which discovery should be placed first in the timeline?
   a. Schleiden concluded plants are made of cells
   b. Hooke observed cork cells
   c. Virchow stated cells come from pre-existing cells
   d. Leeuwenhoek observed microorganisms

What I Need to Know

This supplementary module is designed to help you understand the historical development of **Cell Theory**, one of the most important foundations in biology. In this lesson, you will explore how scientific knowledge develops through the observations and discoveries of different scientists over time.

Lesson 1 The Cell Theory

Cell theory is one of the fundamental principles in biology, it explains the basic structure and function of all living organisms. Cell theory lies in its role as the foundation for studying life. Understanding cell theory helps explain how organisms grow, develop, and carry out the processes necessary for survival.

After going through this supplementary module, you are expected to:

1. analyze the development of cell theory thru historical analysis of scientific information;
2. organize historical timeline of discoveries of cell theory; and
3. explain how individual discoveries contributed to cell theory.

What's In

Activity 1: "My Mystery Scientist"

Directions: There are four (4) scientists having their conversation about their discovery. Your task is to find the mystery scientists and match it to their corresponding discoveries. Use the clues provided below.

- "I found out that all plants are composed of cells."
- "I first used a microscope to observe cork and named the tiny compartments 'cells.'"
- "I proposed that cells came from other existing cells."
- I discovered microorganisms using a powerful single-lens

What's New

Activity 2: "Secondary Source Investigation"

Directions: Read the passage about cell theory and complete the historical analysis table.

The cell theory is one of the most important ideas in biology because it explains that cells are the basic units of life. This theory was developed through the discoveries and observations of several scientists who studied microscopic structures using early microscopes.

In 1665, Robert Hooke observed a thin slice of cork using a compound microscope. He noticed many tiny box-like compartments and called them cells, because they reminded him of small rooms. Although Hooke observed dead cells from plant tissue, his work introduced the concept that living things are made up of small structural units.

Several years later, in 1674 Antonie van Leeuwenhoek improved the microscope and became the first person to observe living microorganisms in pond water and other samples. He called these tiny living organisms "animalcules." His discoveries showed that microscopic life exists and that cells can be living organisms.

In 1838, Matthias Schleiden studied plant tissues and concluded that all plants are composed of cells. His research helped scientists understand that cells are the basic units that make up plant structures.

Later, in 1855, Rudolf Virchow proposed that all cells come from pre-existing cells. This idea explained how cells reproduce and how organisms grow and develop.

Together, the discoveries of these scientists helped establish the cell theory, which states that all living organisms are made of cells, the cell is the basic unit of life, and new cells arise from existing cells.

Source: https://tinyurl.com/58smc7vc

HISTORICAL ANALYSIS OF CELL THEORY				
Scientist	**Year of Discovery**	**Observation or Study Conducted**	**Major Discovery**	**Contribution to Cell Theory**
Robert Hooke				
Antoine Van Leeuwenhoek				
Matthias Schleiden				
Rudolf Virchow				

The Three Main Principles of Cell Theory

1. **All living organisms are made up of one or more cells.** This means that every living thing, from single-celled bacteria to complex organisms like plants and animals, is composed of cells.
2. **The cell is the basic unit of structure and function in living organisms.**

Cells carry out all the processes necessary for life, such as metabolism, growth, and reproduction.

3. **All cells arise from pre-existing cells.** New cells are formed through the division of existing cells.

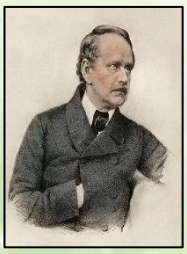
What's More

Activity 3: "Cell Theory Timeline"

Directions: Organize the cell theory timeline. Cut out the picture of each scientist and their discovery and paste them to its corresponding year discovered.

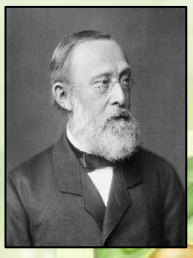
Scientist				
	1665	**1674**	**1838**	**1855**
Discovery				

Discovered living microorganisms (animalcules)



Concluded that all plants are composed of cells

Proposed that new cells arise from existing cells



Discovered small box-like structures and named them "cells"

Guide Questions:

1. What was Robert Hooke's contribution to the study of cells?

   _____

   _____

2. How did Antonie van Leeuwenhoek's discovery improve our understanding of living organisms?

   _____

   _____

3. What conclusion did Matthias Schleiden make about plants?

   _____

   _____

4. What important idea did Rudolf Virchow introduce about the origin of cells?

   _____

   _____

5. Explain how the discoveries of these scientists contributed to the development of the Cell Theory.

   _____

   _____

6. Why is it important to study the historical development of scientific ideas like the Cell Theory?

   _____

   _____

To experience **MORE** interactivity in this module, scan this QR code.

What I've Learned

Directions: Encircle the letter of the correct answer.

1. A researcher studying plant tissues concluded that all plants are composed of cells. Later, scientists expanded this idea to include animals. Which scientist made the original observation about plants?
   a. Matthias Schleiden
   b. Rudolf Virchow
   c. Antonie van Leeuwenhoek
   d. Robert Hooke
2. A teacher asks students why Virchow's statement "Omnis cellula e cellula" (all cells come from cells) is important. Which explanation best demonstrates higher-level understanding?
   a. It explains that cells are very small.
   b. It shows that living organisms grow through cell division.
   c. It proves microscopes are important tools.
   d. It explains why cork has cells.
3. A student says that Hooke and Leeuwenhoek made similar discoveries. Which explanation best evaluates this statement?
   a. They both discovered plant cells only.
   b. Hooke observed dead cork cells while Leeuwenhoek observed living microorganisms.
   c. They both proposed cell theory.
   d. They both studied only animal tissues.
4. A student created the following timeline:
   - Leeuwenhoek observed microorganisms
   - Hooke observed cork cells
   - Virchow proposed that cells come from existing cells

   What is the best critique of this timeline?
   a. The timeline is correct.
   b. Hooke's discovery should come before Leeuwenhoek's discovery.
   c. Virchow's discovery should come before Hooke's discovery.
   d. Leeuwenhoek's discovery should come after Virchow's discovery.
5. A science historian explains that the development of cell theory is an example of cumulative scientific knowledge. What does this mean?
   a. Scientific discoveries occur randomly.
   b. Scientific ideas build upon previous discoveries.
   c. Only one scientist develops scientific theories.
   d. Scientific discoveries are always made at the same time.